(ii) Evaluation of $f^{-1}[f(x)]$, where f is any one of the six trigonometric functions.

(a) If x is in the restricted domain (principal domain) of f, then $f^{-1}[f(x)] = x$.

(b) If x is not in the restricted domain of f, then find x_1 within the restricted domain of f such that $f(x) = f(x_1)$. Now, $f^{-1}[f(x)] = x_1$. For instance,

$$\sin^{-1}(\sin x) = \begin{cases} x & \text{if } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ x_1 & \text{if } x \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \text{where } \sin x = \sin x_1 \text{ and } x_1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]. \end{cases}$$

Property-III (Reciprocal inverse identities)

(i) $\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$, if $x \in \mathbb{R} \setminus (-1, 1)$. (ii) $\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x$, if $x \in \mathbb{R} \setminus (-1, 1)$.

(iii) $\tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x & \text{if } x > 0 \\ -\pi + \cot^{-1} x & \text{if } x < 0. \end{cases}$

Proof

(i) If $x \in \mathbb{R} \setminus (-1, 1)$, then $\frac{1}{x} \in [-1, 1]$ and $x \neq 0$. Thus, $\sin^{-1}\left(\frac{1}{x}\right)$ is well defined.

Let $\sin^{-1}\left(\frac{1}{x}\right) = \theta$. Then, by definition $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$ and $\sin\theta = \frac{1}{x}$.

Thus, $\operatorname{cosec}\theta = x$, which in turn gives $\theta = \operatorname{cosec}^{-1} x$.

Now, $\sin^{-1}\left(\frac{1}{x}\right) = \theta = \operatorname{cosec}^{-1} x$. Thus, $\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$, $x \in \mathbb{R} \setminus (-1, 1)$.

Similarly, other results are proved.

Property-IV (Reflection identities)

(i) $\sin^{-1}(-x) = -\sin^{-1} x$, if $x \in [-1, 1]$.

(ii) $\tan^{-1}(-x) = -\tan^{-1} x$, if $x \in \mathbb{R}$.

(iii) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$, if $|x| \geq 1$ or $x \in \mathbb{R} \setminus (-1, 1)$.

(iv) $\cos^{-1}(-x) = \pi - \cos^{-1} x$, if $x \in [-1, 1]$.

(v) $\sec^{-1}(-x) = \pi - \sec^{-1} x$, if $|x| \geq 1$ or $x \in \mathbb{R} \setminus (-1, 1)$.

(vi) $\cot^{-1}(-x) = \pi - \cot^{-1} x$, if $x \in \mathbb{R}$.

Proof

(i) If $x \in [-1, 1]$, then $-x \in [-1, 1]$. Thus, $\sin^{-1}(-x)$ is well defined

Let $\sin^{-1}(-x) = \theta$. Then $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $\sin\theta = -x$.

Now, $\sin\theta = -x$ gives $x = -\sin\theta = \sin(-\theta)$

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